

Subject continuums: Mathematics

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Primary Years Programme

Subject continuums: Mathematics

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IB mission statement

The International Baccalaureate aims to develop inquiring, knowledgeable and caring young people who help to create a better and more peaceful world through intercultural understanding and respect.

To this end the organization works with schools, governments and international organizations to develop challenging programmes of international education and rigorous assessment.

These programmes encourage students across the world to become active, compassionate and lifelong learners who understand that other people, with their differences, can also be right.



IB learner profile

The aim of all IB programmes is to develop internationally minded people who, recognizing their common humanity and shared guardianship of the planet, help to create a better and more peaceful world.

As IB learners we strive to be:

INQUIRERS

We nurture our curiosity, developing skills for inquiry and research. We know how to learn independently and with others. We learn with enthusiasm and sustain our love of learning throughout life.

KNOWLEDGEABLE

We develop and use conceptual understanding, exploring knowledge across a range of disciplines. We engage with issues and ideas that have local and global significance.

THINKERS

We use critical and creative thinking skills to analyse and take responsible action on complex problems. We exercise initiative in making reasoned, ethical decisions.

COMMUNICATORS

We express ourselves confidently and creatively in more than one language and in many ways. We collaborate effectively, listening carefully to the perspectives of other individuals and groups.

PRINCIPLED

We act with integrity and honesty, with a strong sense of fairness and justice, and with respect for the dignity and rights of people everywhere. We take responsibility for our actions and their consequences.

OPEN-MINDED

We critically appreciate our own cultures and personal histories, as well as the values and traditions of others. We seek and evaluate a range of points of view, and we are willing to grow from the experience.

CARING

We show empathy, compassion and respect. We have a commitment to service, and we act to make a positive difference in the lives of others and in the world around us.

RISK-TAKERS

We approach uncertainty with forethought and determination; we work independently and cooperatively to explore new ideas and innovative strategies. We are resourceful and resilient in the face of challenges and change.

BALANCED

We understand the importance of balancing different aspects of our lives—intellectual, physical, and emotional—to achieve well-being for ourselves and others. We recognize our interdependence with other people and with the world in which we live.

REFLECTIVE

We thoughtfully consider the world and our own ideas and experience. We work to understand our strengths and weaknesses in order to support our learning and personal development.

The IB learner profile represents 10 attributes valued by IB World Schools. We believe these attributes, and others like them, can help individuals and groups become responsible members of local, national and global communities.

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Mathematics strands

Data handling	Data handling includes analysing, making inferences about and drawing conclusions from what we know about the world. Primary and secondary data can be collected, organized, represented and summarized in a variety of ways to highlight similarities, differences and trends. Mathematicians evaluate the chosen format to allow for the bias or distortion of data. Probability, both theoretical and experimental, involves the application of mathematical ideas to describe the level of certainty that something will happen.
Measurement	To measure is to assign a number to a quantity using a chosen tool and unit, reflecting on the degree of accuracy that is necessary for the context. Measurement involves length, area, mass, capacity, money, temperature and time. Since the attributes being measured are continuous, ways must be found to deal with quantities that fall between numbers.
Shape and space	Reasoning about shape and space promotes an understanding of how objects relate to one another in everyday environments. Analysing shapes, identifying properties of shapes and having an awareness of location, position and direction promote spatial reasoning.
Pattern and function	Patterns are regularities that can be observed to help make sense of the world. The repetitive features of patterns can be identified and described as generalizations called “functions”. Making generalizations is foundational to algebraic reasoning. Understanding mathematical properties (i.e. commutative, distributive and associative properties) that stem from regularities in operations with numbers supports the development of <i>computational fluency</i> and <i>computational thinking</i> .
Number	Numbers and operations with numbers can be expressed as symbols and words to represent and describe quantity. Numbers are used to communicate mathematical concepts and skills, as well as interpretations and decisions during problem-solving and investigations. Decomposing and composing numbers in increasingly flexible and efficient ways promote computational fluency.

Overall expectations

Overall expectations				
Strand: Data handling				
Phase 1	Phase 2	Phase 3	Phase 4	
Learners develop an understanding of how the collection and organization of information help people to answer questions and make sense of the world. They ask and answer questions, and develop strategies to collect data. They determine ways to sort objects and organize data into self-selected as well as predetermined categories. They develop strategies for representing quantitative data as graphs, including using symbolic notation such as pictographs, tally marks or objects. Learners respond to questions about the data and discuss the likelihood of events based on their personal experiences.	Learners understand that people collect different types of data to answer questions and to understand relationships. Learners develop strategies, including observations, experiments and interviews, to collect data on relevant issues. They collect, organize, display and interpret data, and understand that information can be organized in different ways for different purposes. Learners discuss and compare the likelihood of events in familiar contexts and make predictions using appropriate mathematical language such as “likely, unlikely, certain and impossible”. Learners make predictions based on personal experiences and simple experimental data.	Learners continue to collect, organize, visualize, analyse and interpret their own data. Learners understand how information can be organized for different purposes. Learners also analyse and interpret external data, developing an understanding of how different graphs highlight different aspects of data more efficiently. They understand that scale can represent different quantities in graphs and that mode as a summary of data tells some stories while obscuring others. Learners use data to make informed decisions on relevant issues. They make the connection that probability is based on experimental events and can be expressed numerically, using increasingly precise mathematical language.	Learners continue to expand their repertoire of strategies for collecting, organizing, visualizing, analysing and interpreting data from a variety of sources, understanding that different graphs highlight different aspects of data more efficiently. They use mode, median, mean and range to summarize a set of data, identifying the stories told and obscured in each. Learners use data to make informed decisions and convincing arguments on relevant issues. Learners understand that probability can be expressed on a scale and that the probability of an event can be predicted theoretically. Learners evaluate the appropriateness of a suggested probability.	

Overall expectations				
Strand: Measurement				
Phase 1	Phase 2	Phase 3	Phase 4	
Learners develop an understanding that objects have attributes that can be measured and compared. They develop an understanding of how measurement involves the comparison of objects and the ordering and sequencing of events. Learners identify, compare and describe attributes of real objects as well as describe and sequence familiar events in their daily routine. They use the language of time in relation to familiar contexts. Learners may measure using non-standard units that differ from person to person and time to time.	Learners continue to measure using non-standard units and develop an understanding that standard units allow for a common language to measure and describe objects and events across contexts. They understand that while estimation is a strategy that can be applied for approximate measurements, particular tools allow objects and events to be measured and described with more accuracy. Learners describe more than one measurement associated with an object, shape or event. Learners begin to recognize that measurements are <i>conserved</i>.	Learners continue to use standard units to measure objects more flexibly and precisely. In particular, they develop their understanding of measuring perimeter, area and volume. They select and use appropriate tools and units of measurement with a focus on accuracy, and describe measures that fall between two numbers on a scale. Learners use and relate standard units associated with length, capacity, mass, time and money measurements. Learners recognize when estimated measurements and exact measurements are used.	Learners identify whether an estimated or exact measurement is appropriate in a situation. Learners measure flexibly and precisely, using a variety of tools and units to measure attributes. Learners understand that a range of procedures exists to measure different attributes of objects and events, for example, the use of formulas for finding area, perimeter and volume. They decide on the level of accuracy required for measuring, using decimal and fraction notation when precise measurements are necessary. To demonstrate their understanding of angles as a measure of rotation, the learners measure and construct angles.	

Overall expectations				
Strand: Shape and space				
Phase 1	Phase 2	Phase 3	Phase 4	
Learners recognize shapes in the real world. They distinguish between 2D and 3D shapes. Learners understand that shapes have characteristics that can be described and compared. They understand and use positional language to describe paths, regions and boundaries of their immediate environment.	Learners continue to work with 2D and 3D shapes, developing the understanding that shapes are classified and named according to their properties. They understand that examples of symmetry and transformations can be found in their immediate environment. Learners interpret, create and use simple directions and specific vocabulary to describe paths, regions, positions and boundaries of their immediate environment. They recognize the different effects of slides and flips.	Learners sort, describe and model regular and irregular polygons, developing an understanding of their properties. They describe and model congruency and similarity in 2D and 3D shapes. Learners continue to develop their understanding of symmetry, and in particular, reflective and rotational symmetry. They understand how geometric shapes and associated vocabulary are useful for representing and describing objects and events in <i>authentic situations</i> .	Learners recognize, name and describe geometric properties in the real world. Learners understand the properties of regular and irregular polyhedra. They understand the properties of 2D shapes and understand that 2D representations of 3D objects can be used to visualize and solve problems in the real world, for example, using drawing and modelling. Learners develop their understanding of the use of scale (ratio) to enlarge and reduce shapes. They apply the language and notation of bearing to describe direction and position. Learners identify location in a coordinate system using ordered pairs.	

Overall expectations				
Strand: Pattern and function				
Phase 1	Phase 2	Phase 3	Phase 4	
Learners identify patterns in personal environments. They identify, describe, extend and create repeating patterns and growing patterns.	Learners continue to identify patterns in personal environments. They identify, describe, extend, create and compare repeating patterns and growing patterns. Learners model patterns in a variety of ways.	Learners identify a broader range of patterns (i.e. complex repeating patterns, growing patterns, shrinking patterns) in the environment. They identify, describe, extend, create and compare a broader range of patterns. Learners model patterns in a variety of ways, make connections between the different representations and identify pattern rules.	Learners continue to identify complex patterns (i.e. complex growing patterns and shrinking patterns) in the environment. They identify, describe, extend, create and compare complex patterns. Learners model complex patterns in a variety of ways, make connections between the different representations and identify pattern rules. Learners use pattern rules and/or generalizations to find the term position and the term value.	

Overall expectations				
Strand: Number				
Phase 1	Phase 2	Phase 3	Phase 4	
Learners understand that numbers tell how much of something there is. Learners recognize that there are different representations for the same number (i.e. 10 can be represented as $5 + 5$). Learners represent numbers concretely or symbolically. Learners rely on benchmark numbers of 5 and 10 when making comparisons between numbers. Learners demonstrate evidence of joining and taking away or comparing in addition and subtraction situations.	Learners develop their understanding of the base ten place value system and model, read, write, estimate, compare and order numbers to hundreds or beyond. Learners describe contexts where numbers to 100 or beyond are found in the environment. Learners identify addition and subtraction situations, and relate addition to subtraction. They understand that addition and subtraction of numbers are facilitated by decomposing and composing numbers in efficient ways, including by place value. Learners describe fractions as part of a set or part of a whole. Learners understand fractions as representations of part-whole relationships and model fractions and use fraction names in various authentic situations. Learners rely on mathematical principles (for example, commutative property, associative property) when performing computations. Learners develop fluency and flexible thinking about numbers, and apply mental mathematics strategies when performing computations.	Learners develop their understanding of the base ten place value system and model, read, write, estimate, compare and order numbers between hundreds and beyond. Learners begin to develop multiplicative thinking. Learners identify multiplication and division situations, and make connections between addition, subtraction, multiplication and division. Learners develop the understanding that fractions and decimals are ways of representing part-whole relationships, and demonstrate this understanding by modelling equivalent fractions and decimal fractions to hundredths and beyond. They model, read, write, compare and order fractions, and use them in authentic situations. Learners continue to develop fluency and flexible thinking about numbers, and apply mental mathematics strategies when performing computations. They select, use and describe a range of strategies to solve problems involving addition, subtraction, multiplication and division. They use estimation strategies to check the reasonableness of their answers.	Learners understand that the base ten place value system extends infinitely in two directions and model, compare, read, write and order numbers to millions or beyond, as well as model integers. Learners describe contexts where numbers, including integers, would be found in the environment. Learners solve ratio and rate problems. They understand that fractions, decimals and percentages are ways of representing part-whole relationships, and work towards modelling, comparing, reading, writing and ordering fractions, decimals and percentages. Learners explain the connections between fractions, decimals and percentages, and represent the same part-whole relationship in different ways. Learners use mental and written strategies to solve problems involving whole numbers, fractions and decimals in authentic situations. They use a range of strategies to evaluate the reasonableness of answers. Learners apply their fluency and flexibility with confidence and accuracy when performing computations	

Strand: Data handling

Example concepts: chance, communication, data, experiment, information, interpretation, notation, organization, probability, quantity, relationship, scale, symbol.

Example conceptual understandings				
Strand: Data handling				
Phase 1	Phase 2	Phase 3	Phase 4	
Collecting information helps people make sense of the world around them.	Information can be expressed as organized data to facilitate the understanding of relationships.	Appropriate questions can lead to the intentional collection of data. Following the collection, further questions may emerge.	Presented purposefully, accurately and effectively, data can be used for valid interpretation and communication.	
Organizing objects and events helps mathematicians make sense of mathematical situations.	Organizing objects and events in different ways reveals different relationships.	Data can be collected, organized, displayed and analysed in different ways.		
Events in daily life involve chance.	Some events in daily life are more likely to happen than others.	Different graph forms highlight different aspects of data more efficiently.	Analysing data supports people in discovering patterns and making predictions.	
		Probability can be expressed numerically—as a fraction, decimal or percentage.	Range, mode, median and mean can be used to summarize and analyse statistical data.	
			Probability can be represented on a scale of 0–1 or 0%–100%.	
			The probability of an event can be predicted theoretically.	

Example learning outcomes				
Strand: Data handling				
Phase 1	Phase 2	Phase 3	Phase 4	
Learners: - sort and describe sets of loose parts using different attributes - obtain information about themselves and others, their surroundings and their interests in different ways - represent information related to their own inquiries, for example, through pictographs or tally marks - create graphs using objects and people - discuss chance in daily events (impossible, maybe, certain).	Learners: - organize sets by one or more attributes - represent the relationship between objects in sets using concrete objects and drawings such as tree, Venn and Carroll diagrams - collect and record information about themselves and others, their surroundings and their interests in different ways - collect and represent data related to their own inquiries in different types of graphs such as pictographs and bar graphs - collect, display and interpret data for the purpose of answering questions (for example, inventories and surveys) - create graphs, such as pictographs and bar graphs, of real objects and interpret data by comparing quantities using the vocabulary of more, fewer, less than, greater than - identify and describe examples of chance in daily events using the vocabulary of impossible, less likely, maybe, most likely, certain.	Learners: - design surveys and systematically collect, organize and display data in a variety of graphs - collect, display and interpret data using appropriate graphs such as bar graphs and line graphs - utilize a <i>database</i> to answer questions and solve problems - summarize a set of data through finding the mode - identify, read and interpret range and scale on graphs - use probability to determine mathematically fair and unfair games, and to explain possible outcomes - express probability using simple fractions, including the use of tree diagrams.	Learners: - set up spreadsheets, use simple formulas to manipulate data and create graphs - design surveys and systematically collect, record, organize and display data - compare and evaluate different displays of data for a specific purpose, for example, a double bar graph may be more effective in certain situations - identify, describe and explain the range, mode, median and mean in a set of data - given a measure of central tendency, predict what the data may have been - determine the theoretical probability of an event and explain why it might differ from experimental probability - analyse a set of data and make generalizations and predictions - express probabilities using scale (0–1) or percentage (0%–100%).	

Strand: Measurement

Example concepts: attributes, comparison, conversion, estimation, formula, order, sequence, unit.

Example conceptual understandings				
Strand: Measurement				
Phase 1	Phase 2	Phase 3	Phase 4	
Measurement involves comparing objects and events.	Standard units allow us to have a common language to identify, compare, order and sequence objects and events.	Objects and events have multiple attributes that can be measured using appropriate tools.	Conversion of units and measurements allows us to make sense of the world we live in.	
Objects have attributes that can be measured using non-standard units.	Mathematicians choose from a range of tools to measure the attributes of objects and events.	The same object or event can be described using different measurements and tools.	The use of standard measurement units simplifies communication about the size of the objects.	
Events can be ordered and sequenced.	Estimation allows mathematicians to measure with different levels of accuracy.	Different measurements of the same object are not always related (i.e. an object could be really big, but it might not weigh much).	There is always a smaller, more precise unit of measure.	
	The same object can be described using different measurements.	All measurements are represented using a numerical value and a unit.	Most measurements can be determined using a combination of units.	
		Connections between the place value system and units of measurement exist.	Accuracy of measurements depends on the situation and the precision of the tool.	
		Measurement is a comparison (for example, the length of a pencil can be compared to a unit in centimetres).		

Example learning outcomes				
Strand: Measurement				
Phase 1	Phase 2	Phase 3	Phase 4	
Learners: • identify, describe and compare attributes of real objects (for example, longer, shorter, heavier, empty, full, hotter, colder) • identify, describe and sequence events in daily routines (for example, before, after, bedtime, today, tomorrow) • compare the length, mass and capacity of objects using non-standard units • describe observations about events and objects in authentic situations • use non-standard units of measurement fairly to solve problems in authentic situations involving length, mass and capacity.	Learners: • use standard units to estimate and measure (for example, length, mass, money, time, temperature, capacity) • select an appropriate tool to measure length, mass, time, capacity and temperature • use calendars to determine the date, and to identify and sequence days of the week and months of the year • make connections between universal units of measure (for example, years, months, days, hours, minutes and seconds) • read and write time to the hour, half hour and quarter hour • estimate and compare lengths of time: second, minute • measure real objects in authentic situations • use measures of time to assist with problem-solving in authentic situations.	Learners: • use standard units of measurement to solve problems in authentic situations involving perimeter, area and volume • select appropriate tools and units of measurement to determine an accurate measurement • use timelines in units of inquiry and other authentic situations • determine a measurement using different units • convert measurements • collect, display and interpret data for the purpose of using standard units of measurement to solve problems in authentic situations involving length, mass, capacity, money and temperature.	Learners: • use decimal and fractional notation when measuring for greater precision (for example, 3.2 cm, 1.47 kg, 1.5 kilometres) • read and interpret scales on a range of measuring instruments • measure and construct angles in degrees using a protractor • make unit conversions within a system of measurement (i.e. between mm, cm and m) • describe how capacity and volume are related, and how they are different • select and use appropriate tools and units of measurement to solve problems in authentic situations • determine and justify the level of accuracy required to solve authentic problems involving measurement • use timetables and schedules (12-hour and 24-hour clocks) in authentic situations, including determining times worldwide.	

Strand: Shape and space

Example concepts: comparison, model, position, property, scale, shape, space, symmetry, transformation.

Example conceptual understandings				
Strand: Shape and space				
Phase 1	Phase 2	Phase 3	Phase 4	
Shapes can be described and organized according to their properties.	Shapes are classified and named according to their properties.	A property is an attribute that applies to all shapes in a specific group (for example, a group of rectangles has the following properties: four straight sides and four right angles).	Manipulation of shape and space takes place for a particular purpose.	
There are many different ways to represent a given shape.	Some shapes are made up of parts that repeat in some way.	Shapes can be classified by their attributes (for example, size, shape, colour) or their properties.	Geometric tools and methods can be used to solve problems relating to shape and space.	
Objects in the immediate environment have a position in space that can be described in relation to one another.	Specific vocabulary can be used to describe an object's position in space.	Any shape can be created by combining or dissecting other shapes.	Geometric ideas and relationships can be used to solve problems in other areas of mathematics and in authentic situations.	
	Transformations of shapes (slides, flips and turns) can be found in natural and human-made environments.	Shapes can be transformed in different ways without impacting their properties.	Points reference position on a cartesian plane.	
	Geometric shapes and vocabulary are useful for representing and describing objects and events in authentic situations.	All regular polygons are convex shapes that have all sides equal in length and all angles equal in size.		

Example learning outcomes				
Strand: Shape and space				
Phase 1	Phase 2	Phase 3	Phase 4	
Learners: - describe and compare 2D and 3D shapes used in construction based on their characteristics - use common language in games and play to describe position and direction, for example, inside, above, next to, in front of, down - sort 2D and 3D shapes based on their characteristics.	Learners: - describe the relationships among and between 2D and 3D shapes - create 2D and 3D shapes using different materials by putting together and/or taking apart other shapes - identify examples of symmetry and transformations in their immediate environment - demonstrate that geometric shapes are useful for representing meaningful authentic situations - select and use shapes to fill a space without gaps or overlaps when completing a geometric puzzle or pattern - sort, describe and label 2D and 3D shapes - create and describe symmetrical and tessellating patterns - identify lines of reflective symmetry - represent ideas about the real world using geometric vocabulary and symbols, for example, through oral description, drawing, modelling, labelling - interpret and create simple directions, describing paths, regions, positions and boundaries of their immediate environment.	Learners: - use mathematical terms to describe the properties and sort regular and irregular polygons - identify congruent and similar shapes - use lines and axes of reflective and rotational symmetry to assist with the construction of shapes - use coordinates to identify locations on a grid - describe, analyse and classify 2D and 3D shapes using geometric vocabulary - create models of 2D and 3D shapes (for example, nets, concrete models) - measure and analyse angles by comparing and describing rotations: whole turn; half turn; quarter turn; north, south, east and west on a compass - describe and/or represent mental images of objects, patterns and paths - recognize and explain symmetrical patterns, including tessellations, in the environment - apply knowledge of transformations to problem-solving situations.	Learners: - use mathematical terms to identify and compare shapes - describe the properties of regular and irregular polyhedra in order to compare shapes - use the properties of circles to construct a circle to the specified size - use systems for describing position and direction - use 2D representations of 3D objects to visualize and solve problems, for example, using drawings or models - describe lines and angles using geometric vocabulary - identify and use scale (ratios) to enlarge and reduce shapes - create and model how a 2D net converts into a 3D shape and vice versa - predict how shapes will look after two or more shapes have been combined, or if a shape has been dissected - relate the number of vertices of a polygon to the number of sides - explore the use of geometric ideas and relationships to solve problems in other areas of mathematics - use geometric vocabulary when describing shape and space in mathematical situations and beyond - plot points and describe paths on coordinate grids.	

Strand: Pattern and function

Example concepts: balance, connection, equation, expression, function, pattern, relationship, variable.

Example conceptual understandings				
Strand: Pattern and function				
Phase 1	Phase 2	Phase 3	Phase 4	
Patterns occur in everyday situations.	Whole numbers exhibit patterns and relationships that can be observed and described.	Patterns can be represented in a variety of different ways (for example, graph, pictorial, table, algebraic expression).	Identifying a pattern and making a generalization reflect algebraic reasoning.	
Patterns repeat and grow.	Patterns can be represented using numbers and other symbols.	Analysing patterns supports mathematicians in making generalizations and predictions.	An equation represents balance.	
	Patterns repeat, grow and shrink.	Functions are relationships between two variables.	The representation of a pattern tells us different things about the pattern (for example, $3a + 2$ tells us that 2 is constant).	
		An equation connects two equivalent expressions.	Variables can be used to describe quantities that can be changed.	
		There are an infinite number of ways to represent an algebraic expression (for example, $3a + 2a$ is the same as $a + 4a$).	The ideas developed for whole-number computations apply to variables.	
			The same algebraic expression can be used to describe different <i>authentic contexts</i> .	
			Patterns can often be generalized using algebraic expressions, equations or functions.	

Example learning outcomes				
Strand: Pattern and function				
Phase 1	Phase 2	Phase 3	Phase 4	
Learners: - identify patterns that can be found in everyday situations (for example, sounds, actions, objects, nature) - describe patterns in various ways (for example, words, drawings, symbols, materials, actions, numbers and movements) - extend and create repeating and growing patterns.	Learners: - explore, identify and describe patterns that can be found in numbers (for example, odd and even numbers, skip counting) - explain the inverse relationship between addition and subtraction - represent patterns in a variety of playful ways (for example, using words, drawings, symbols, materials, actions, numbers and movements) - extend and create patterns in numbers (for example, odd and even numbers, skip counting) - use number patterns to represent and interpret authentic situations.	Learners: - identify patterns and make generalizations - demonstrate that multiplication can be described as repeated addition and that division can be described as repeated subtraction - show the inverse relationship between multiplication and division - use the order of operations to demonstrate the associative, distributive and commutative properties of multiplication - explore computational relationships using patterning (for example, multiplying by 2, 20, 200) - describe that patterns exist in the place value system - represent rules or generalizations for patterns in different ways (for example, 2, 4, 6, 8, ... begin with 2 and add 2 each time or "2 x position number") - select appropriate methods for representing patterns - use generalizations to make predictions and solve problems.	Learners: - demonstrate that patterns can be generalized by a rule - represent exponents as repeated multiplication - explore the inverse relationship between exponents and roots - demonstrate that patterns can be represented, analysed and generalized - represent the rule (or generalization) of a pattern to reflect the relationship between the position of the term and the value of the term - recognize the connections between the different representations of the same pattern - use generalizations to solve problems - apply knowledge of the order of operations to solve equations and evaluate expressions - solve equations and simplify expressions.	

Strand: Number

Example concepts: comparison, computation, decimal, fraction, number, operation, order, quantity, ratio, representation, scale, symbol, unit, value.

Example conceptual understandings				
Strand: Number				
Phase 1	Phase 2	Phase 3	Phase 4	
Numbers can be represented in many different ways.	The base ten place value system is used to represent numbers and number relationships.	There are patterns in the base ten place value system and the position of a digit determines its value.	The base ten place value system extends infinitely in two directions.	
Numbers can represent quantities.	Fractions are ways of representing whole-part relationships.	The base ten place value system can be used to describe, compare, order and compute with numbers.	Part-whole relationships can be represented as fractions, decimals and percentages.	
Numbers can be used in many ways for different purposes.	The operations of addition and subtraction are related to each other, and are used to process information to solve problems.	Fractions can represent parts of a whole and parts of a set.	Whole numbers can be written as fractions.	
A whole group of objects can be split into two or more smaller parts in different ways.	Number operations can be modelled in a variety of ways.	The operations of addition, subtraction, multiplication and division are interconnected, and can be used to perform computations in mathematical situations.	Different representations of a number can be more useful in some situations (for example, $\frac{1}{3}$ of an hour is more useful than 0.3333).	
Making connections between numbers, and experiences with numbers, can help develop number sense.	Mathematical principles (for example, commutative property, associative property) help us play with numbers more flexibly.	There are multiple approaches to computation which vary in efficiency.	There are an infinite number of fractions between any two fractions (for example, $\frac{3}{5}$ and $\frac{4}{5}$).	
Conservation of number keeps the quantity of objects constant regardless of their position and distance from one another.	There are many mental methods that can be applied for exact and approximate computations.	Estimations rely on applying an understanding of place value or number sense.		
Quantities change as amounts are added and taken away.		Equivalent fractions describe the same amount using different fractional parts.		

Example learning outcomes				
Strand: Number				
Phase 1	Phase 2	Phase 3	Phase 4	
Learners: - demonstrate one-to-one correspondence in different ways such as clapping and counting objects, recognizing that the order in which objects are counted does not matter - recognize that the counting sequence (1, 2, 3 ...) stays the same - recognize and use one-digit and two-digit numbers in a variety of different situations - explain that, for a set of objects, the number name of the last object counted describes the quantity of the whole set - represent numbers by combining and partitioning - notice that the number of objects in different positions and orientations stays the same - compare the magnitude of whole numbers, identifying which group of objects is more or less - recognize groups of zero to five objects without counting (<i>subitize</i>)	Learners: - model numbers in diverse ways to hundreds or beyond using the place value system - read, write, compare and order ordinal and cardinal numbers - explain and evaluate mental, drawn and written strategies for adding and subtracting two-digit numbers - combine pairs of number to make 10 and use doubling to combine numbers - use strategies to evaluate how reasonable an answer is - model addition and subtraction of whole numbers in different ways - select an efficient method and tool for solving a problem - estimate quantities to 100 or beyond - estimate sum and difference in authentic situations - model simple fraction relationships.	Learners: - model, read, write, compare and order whole numbers to tens of thousands or beyond using the place value system - use the base ten place value system to reflect on the magnitude of the number - model equivalent fractions (for example, fraction strips, number lines) - represent the same number in a variety of ways (for example, decimal and fraction) - describe the role of the numerator and denominator - model, order and compare decimal numbers represented in different ways - model multiplication and division of whole numbers in different ways - describe the components of multiplication and division statements - describe the role of factors, products, divisors, dividends and quotients - model addition and subtraction of fractions with related denominators - model addition and subtraction of decimals - develop mental maths strategies when adding, subtracting, multiplying and dividing to develop fluency - read, write, compare and order fractions to hundredths or beyond	Learners: - model, read, write, compare and order whole numbers up to millions or beyond - use whole numbers up to millions or beyond in authentic situations - create numbers of various magnitude given conditions - model, read and write ratios and rates - use ratios and rates in authentic situations - model, read and write integers in appropriate contexts, and model the addition and subtraction of integers - model, read and write exponents and square roots - model improper fractions and mixed numbers - convert improper fractions to mixed numbers and vice versa in authentic situations - simplify fractions in mental and written form - model, read, write, compare and order decimal fractions to thousandths or beyond - model, read, write, compare and order percentages - convert between fractions, decimals and percentages	

Example learning outcomes				
Strand: Number				
Phase 1	Phase 2	Phase 3	Phase 4	
- model part-whole relationships in various ways - count forwards and backwards in ones from different numbers to and from 20 - locate a number from 0 to 20 on a number track, using logic to explain thinking (for example, 9 is near the middle) - use words and numerals to represent quantities in relevant contexts such as games, friendships and play - estimate a quantity of objects (up to 20).		- multiply and divide using different strategies (for example, area model for multiplication, partial products, partial quotients) - make connections between the different strategies used when multiplying and dividing - identify fractions in authentic situations - estimate product and quotient in authentic situations, including fractions and decimals - use strategies to evaluate the reasonableness of answers - perform computations with numbers in different forms, including fractions and decimals - select and use an appropriate sequence of operations to solve word problems and equations.	- use fractions, decimals and percentages interchangeably in authentic situations - model addition, subtraction, multiplication and division of fractions - recognize that strategies used to add whole numbers can be applied to adding fractions - model addition, subtraction, multiplication and division of decimals - connect division to fractions - read and write integers in appropriate contexts - use integers in authentic situations - identify when a specific representation of a number is more useful - perform computational estimations - select an efficient method for solving a problem: mental estimation, mental computation, written algorithms, by using a calculator - use mental and written strategies for adding, subtracting, multiplying and dividing fractions and decimals in authentic situations, and use strategies to evaluate the reasonableness of answers - estimate and make approximations in authentic situations involving fractions, decimals and percentages - select and use an appropriate sequence of operations to solve problems in authentic situations.	

Mathematics glossary

Authentic context	The broader setting in which a mathematical problem or scenario is situated. It may include a particular environment, location, or cultural context.
Authentic situations	Relevant life scenarios that require the application of mathematical concepts and skills.
Computational fluency	The ability to compute efficiently, flexibly and accurately.
Computational thinking	The thought processes involved in formulating a problem and expressing its solution in precise steps that a person or a machine can effectively carry out. For example, exploring coding to determine how to move a robot in different directions.
Conserve	In mathematical terms, the amount stays the same regardless of the arrangement.
Database	A collection of data, where the data can be displayed in many forms; the data can be changed at any time; a spreadsheet is a type of database.
Model	Using concrete materials to represent numbers or number operations, for example, the use of pattern blocks or fraction pieces to represent fractions, and the use of base ten blocks to represent number operations.
Subitize	To determine the number of objects at a glance, without counting.